Inorganic and Physical Chemistry

Science

May, 2026

setu.ie

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Inorganic and Physical Chemistry (A11244)

Short Title: Inorganic & Physical Chemistry
Faculty Science and Computing
Department Science
Cluster Chemistry
Author EOWENS
Credits 5 **Level:** Intermediate

Description of Module / Aims

This module will give the student an in-depth knowledge of metal to ligand bonding in inorganic materials, and demonstrates how the bonding plays an important role in the properties of these complexes. The electronic absorption spectra and infra-red (IR) spectra of inorganic materials will also be discussed. The determination of crystal structure of compounds and its importance in pharmaceutical applications will be covered. The importance of polymorphism, particle size and surface characateristics of the materials will be studied. The role of kinetics in complex reactions, surface and interfacial science, adsorption isotherms will also be reviewed.

Programmes

CHEM-0030	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	3 / 5 / M
CHEM-0030	BSc in Pharmaceutical Science (WD_SPHSC_D)	3 / 5 / M

Indicative Content

- Bonding in inorganic materials, molecular oribital theory of hetero nuclear molecules, crystal field theory of bonding, electroneutrality principle, ionic bonding, back bonding, sigma donor interactions
- Magnetic properties of inorganic materials, including diamagnetism, paramagnetism, ferromagnetism, spin and orbital contributions to magnetic moment
- Electronic absorption spectra, infra-red (IR) spectroscopy of metal carbonyls
- Isomerisation and the various types of isomers that may occur in inorganic molecules; crystal structures of materials, x-ray diffraction and crystallography; polymorphism and its importance in the pharmaceutical industry
- Surface characterisation of materials including scanning electron microscopy (SEM), scanning tunneling micrcoscopy (STM), atomic force microscopy (AFM) and surface probe microscopy
- Kinetics of complex reactions and their significance to the pharmaceutical and polymer science
- Properties of surfaces, surfactants and interfacial phenomena (including the kinetics and thermodynamics)
- Adsorption Isotherms (Langmuir, Freundlich and BET)

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Apply the various bonding theories for inorganic complexes, and interpret spectroscopic and magnetic properties of these complexes.
2. Demonstrate safely the synthesis, characterisation and structural analysis of inorganic compounds.
3. Appraise the surface characterisation analysis of materials.
4. Demonstrate the structure of a surfactant, uses of surfactants, surfactant aggregation and the driving forces for aggregation.
5. Calculate the critical micelle concentration (CMC), surface excess concentration and relevant thermodynamic parameters.
6. Determine the rate law of complex reactions using steady state approximations, pseudo kinetics, relative rates, relaxation techniques and pre-equilibria conditions. Methodologies for fast reactions.
7. Differentiate between adsorption isotherms (Langmuir, Freundlich and BET), their uses and applications.
8. Complete, safely, and report on a range of practical laboratory experiments that will reinforce the theoretical aspects of this module.

Learning and Teaching Methods

- Lectures.
- Library work.
- Assignments.
- Set examples.
- Laboratory-based practicals.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4,6
Continuous Assessment	50%	
Practical	50%	5,7,8

Assessment Criteria

<40%: Little or no demonstration of an understanding of the fundamentals of inorganic and physical chemistry. Inability to carry out numerical calculations and lack of competence with regard to associated laboratory skills, practices and instrumentation.

40%-49%: Will be able to show a limited understanding of the fundamentals of inorganic and physical chemistry, but may have some difficulty with applying this theory. Will display some ability with regard to associated laboratory skills, practices and instrumentation.

50%-59%: As well as the above, the student will be to show reasonable understanding of the fundamentals of inorganic and physical chemistry. Will display reasonable competence with regards to associated laboratory skills, practices and instrumentation and be able to correctly interpret characterisation of materials.

60%-69%: In addition, the student will demonstrate a more detailed knowledge of inorganic and physical chemistry and show ability to analyse problems in this area. Will display very good competence with regards to associated laboratory skills, practices and instrumentation.

70%-100%: All of the above to an excellent level. In addition, the learner will be able to demonstrate comprehensive knowledge and understanding of inorganic and physical chemistry, excellent reasoning skills and well thought out arguments. Will display advanced competence and independence with regard to associated laboratory skills, practices and instrumentation.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	36	
Independent Learning	63	

Essential Material

- Atkins, P. and J. De Paula. _Elements of Physical Chemistry_. 6th ed. UK: Oxford University Press, 2013.
- Brown, T., H. Lemay, B. Burnstein and C. Murphy. _Chemistry: The Central Science_. 3rd ed. Australia: Pearson, 2013.
- Cotton, F., G. Wilkinson and P. Gaus. _Basic Inorganic Chemistr__y_. 3rd ed. New York: Wiley, 1995.
- Miesslar, G. and D. Tarr. _Inorganic Chemistry_. 3rd ed. New Jersey: Prentice Hall Inc., 2004.

Supplementary Material

- Kettle, S. _Physical Inorganic Chemistry: A Coordination Chemistry Approach_. New York: University Science Books, 1996.
- Lifshin, E. _X-ray Characterization of Materials_. Weinheim: Wiley Verlag GmBH, 1999.
- Rayner-Canham, G. _Descriptive Inorganic Chemistry_. New York: W.H. Freeman & Co., 1996.
- Rosen, M. and J. Kunjappu. _Surfactants and Interfacial Chemistry_. 4th ed. USA: John Wiley and Sons, 2012.
- Woodruff, D., T. Delchar and D. Clarke. _Modern Techniques of Surface Science_. 2nd ed. Cambridge: University Press , 1994.

Requested Resources

- Room Type: Chemistry Lab